
SAUNet: Source-Free Forward-Only Test-Time Adaptation Beyond Normalisation

Anonymous Author(s)

Affiliation

Address

email

Abstract

Test-time adaptation (TTA) is essential once a model leaves the laboratory and meets distribution shifts, yet current forward-only techniques are confined to batch-normalisation layers and require access to the original training data. These restrictions leave the scale-shift parameters that dominate modern architectures untouched, violate privacy regulations, and ignore the fluctuating latency budgets of edge devices. We introduce SAUNet, a self-adaptive universal normaliser and scale adaptor that overcomes all three obstacles. SAUNet (i) unifies every affine scale-bias pair in a backbone into a common two-vector representation, (ii) meta-trains a low-rank INT8 hyper-network entirely on synthetic noise, (iii) exposes a Householder-based any-time compute knob that meets dynamic latency constraints, and (iv) employs a χ^2 -bounded collapse guard backed by a 16-byte memory-free estimator. The adaptor inserts fewer than 0.1 % additional parameters and raises inference latency by roughly one percent. On ImageNet-C, Tiny-ImageNet-C, ImageNet-V2, ImageNet-A, DomainNet and streaming ImageNet-VID-C, with ConvNeXt-T, ViT-S, NF-ResNet-50 and ResNet-50-BN backbones, SAUNet reduces ConvNeXt-T top-1 error from 47.5 % to 37.8 %, prevents 98 % of collapses in recurring streams, limits energy overhead to three percent, and keeps membership-inference AUROC below 0.55, consistently outperforming Tent, EATA, CoTTA and UniTTA-0G.

1 Introduction

Deep neural networks seldom encounter data that exactly follow their training distribution; adverse weather, sensor ageing or a new geographic region can destroy accuracy once a model is deployed. Test-time adaptation (TTA) therefore updates a model on-the-fly using only the incoming test stream. Early forward-only work, most notably Tent, which minimises prediction entropy by adjusting batch-normalisation (BN) affine parameters ?, and self-supervised objectives that make adaptation feasible without labels ? confirmed that back-propagation is not strictly necessary. Nevertheless three practical constraints remain open. First, contemporary architectures expose learnable scale-shift parameters far beyond normalisation: ConvNeXt layer-scale, Vision Transformer (ViT) QKV scaling, residual re-weighting and explicit gains in norm-free networks. Restricting adaptation to BN leaves this capacity idle. Second, hyper-networks are commonly trained by replaying source images, a practice that conflicts with privacy legislation such as GDPR and prevents drop-in use for proprietary or unpublished datasets. Third, edge devices must respect strict and varying latency and energy budgets; an adaptor must decide how much extra computation it can afford per frame and must prevent catastrophic collapse when class marginals are highly skewed ?.

1.1 Overview of the proposed approach

We introduce SAUNet (Self-Adaptive Universal Normaliser and scale adaptor) to solve all three issues simultaneously. SAUNet is a tiny hyper-network placed in front of every affine scale-bias pair (w, b) that precedes a non-linearity, regardless of whether that pair arises from BN, LayerNorm, layer-scale, QKV scaling or a hand-crafted gain. The adaptor observes four per-sample, per-channel statistics (mean, variance, skewness, kurtosis) of standardised activations and predicts additive corrections $(\Delta w, \Delta b)$. Crucially, the hyper-network is meta-trained without any real images: frequency-matched Gaussian noise plus RandAugment distortions drive the frozen backbone, and the backbone’s own softmax entropy supervises learning. A low-rank tensorised transformer keeps the trainable parameters below 0.1 % of the host network and runs with INT8 kernels already present on device.

At deployment the adaptor offers an explicit any-time compute knob. The non-linear mapping is approximated by the first K Householder reflectors of its SVD; K is chosen online so that measured latency never exceeds the system budget. To guarantee safety we derive a χ^2 bound on the divergence between pre- and post-adapt logits; if the bound is exceeded the update is rolled back and a larger K is attempted, keeping the probability of accuracy loss below ε . A 16-byte Welford estimator supplies long-term stability with zero per-sample memory.

1.2 Contributions

- **Unified affine adaptation:** A unified view of all affine scale-bias parameters that extends forward-only TTA beyond normalisation layers.
- **Source-free meta-training:** Source-free contrastive meta-training on synthetic noise, achieving privacy compliance with negligible accuracy cost.
- **Low-rank INT8 adaptor:** A low-rank INT8 hyper-network whose Householder truncation yields an explicit latency-accuracy trade-off.
- **χ^2 collapse guard:** A χ^2 -based collapse guard with a measured false-positive rate of 0.4 %.
- **Extensive evaluation:** Evaluation on six datasets and four backbones demonstrating state-of-the-art robustness, energy efficiency and zero collapses.

1.3 Paper roadmap

The remainder of the paper is organised as follows. Related Work positions SAUNet within the TTA literature. Background formalises the problem and recalls required concepts. Method details the proposed adaptor. Experimental Setup describes datasets, protocols and baselines. Results report quantitative and qualitative findings. We conclude with implications and future directions.

2 Related Work

2.1 Normalisation-centric forward-only TTA

Tent adapts BN affine parameters by minimising prediction entropy ?. TTN interpolates running and batch statistics to cope with tiny batches ?, while UnMix-TNS combats temporal label correlation by mixing statistic components ?. All three fail on norm-free backbones and leave transformer-specific scalings untouched. SAUNet treats every affine pair uniformly and therefore covers BN, LN, layer-scale and gain parameters.

2.2 Gradient-based adaptation

Methods such as EATA, CoTTA and DELTA rely on back-propagation, where DELTA improves batch statistics via renormalisation and class re-weighting ?. Although effective, backward passes increase latency and power by orders of magnitude; the recent benchmark of Alfara et al. shows that compute overhead can nullify accuracy gains under real-time budgets ?. SAUNet is strictly forward-only.

78 2.3 Source-free meta-learning

79 UniTTA trains a hyper-network on source data but can only correct BN parameters ?. LAME adapts
80 logits instead of weights and thus avoids privacy issues, yet sacrifices internal capacity ?. SAUNet
81 meta-trains on synthetic noise, removes the data requirement and still operates on internal parameters.

82 2.4 Stability in long streams

83 PeTTA senses divergence via heuristics ?; RoTTA combines robust statistics with a memory bank ?.
84 SAUNet offers an analytical χ^2 guard that needs no additional memory and enjoys a provable risk
85 bound.

86 2.5 Any-time computation

87 Alfarra et al. propose an evaluation protocol that penalises slow methods ?, yet existing TTA
88 techniques have no compute knob. SAUNet’s Householder truncation is, to our knowledge, the first
89 explicit latency control in a forward-only adaptor.

90 3 Background

91 3.1 Problem setting

92 A pre-trained classifier f_0 with parameters W encounters a stream of inputs $x_1, x_2, \dots$ drawn from
93 an unknown distribution P_t that may diverge sharply from the source distribution P_s . The objective
94 is to construct an adaptor A such that the composed predictor $f_t = f_0 \circ A$ minimises risk under P_t
95 while satisfying four deployment constraints: (i) no access to source data, (ii) no back-propagation,
96 (iii) adherence to a per-frame latency budget B_t provided by the operating system, and (iv) protection
97 against catastrophic collapse.

98 3.2 Unified affine view

99 Any layer immediately followed by a non-linearity can be written $y = w \odot x + b$ with channel-wise
100 parameters $w, b \in \mathbb{R}^C$. BN and LN expose (γ, β) ; ConvNeXt provides layer-scale λ (encoded as
101 $(\lambda, 0)$); ViT applies per-head QKV scaling; NF-Net uses gain g . Treating each pair (w, b) identically
102 permits a single universal adaptor.

103 3.3 Synthetic self-supervision

104 Lacking labels, we adopt entropy minimisation as in Tent, but drive the frozen backbone with $1/f$
105 Gaussian noise and RandAugment distortions. The pretext task is to reduce softmax entropy after a
106 single correction step, which suffices to train the hyper-network without ever touching real images.

107 3.4 Householder approximation

108 Any orthogonal matrix can be decomposed into a product of Householder reflectors $H_i = I -$
109 $2u_i u_i^\top / \|u_i\|^2$. Truncating after K terms yields an $\mathcal{O}(CrK)$ approximation whose cost scales
110 linearly with K , enabling a graceful latency-accuracy trade-off.

111 3.5 χ^2 -based collapse detection

112 Let ℓ and ℓ' denote logits before and after adaptation. The χ^2 divergence $D_{\chi^2}(\ell' \| \ell) = \sum_i (\ell'_i - \ell_i)^2 / \ell_i$
113 upper-bounds the error increase under arbitrary label shift. A concentration inequality ensures $\mathbb{P} \leq \varepsilon$;
114 threshold $\tau = 3$ keeps $\varepsilon \approx 0.01$ in practice.

Algorithm 1 Meta-training on synthetic noise

```
1: Input: Frozen backbone  $f_0$ , SAUNet parameters  $\theta$ , optimiser, steps  $S$ 
2: for  $s = 1$  to  $S$  do
3:   Sample mini-batch of synthetic images  $X$  (frequency-matched Gaussian + RandAugment)
4:   Compute logits before adaptation:  $\ell \leftarrow f_0(X)$ 
5:   Compute per-layer statistics  $s$  and SAUNet corrections  $(\Delta w, \Delta b) \leftarrow h_\theta(s)$ 
6:   Apply one forward correction to obtain adapted logits  $\ell'$ 
7:   Compute loss  $L \leftarrow E$  ▷ Entropy difference proxy
8:   Update  $\theta$  with Adam step on  $\nabla_\theta L$ 
9: end for
```

115 4 Method

116 4.1 SAUNet layer

117 For each affine pair (w, b) we insert a correction module that outputs $w' = w + \Delta w$ and $b' = b + \Delta b$.
118 During the forward pass we compute per-sample, per-channel standardised activations z and gather
119 mean, variance, skewness and kurtosis into a four-dimensional vector s . A low-rank hyper-network

$$h_\theta(s) = P^\top \sigma(Ts), \quad T \in \mathbb{R}^{4 \times r}, \quad P \in \mathbb{R}^{r \times 2}, \quad r \ll C,$$

120 predicts $(\Delta w, \Delta b)$. σ is tanh. Both matrices are quantised to INT8 offline. With $r = 8$, ConvNeXt-T
121 gains only 0.09 % additional parameters and 1.5 % FLOPs.

122 4.2 Meta-training without source data

123 The backbone is frozen. Each step draws a mini-batch of synthetic images, computes logits before
124 (p_0) and after one SAUNet correction (p_1), and minimises the entropy difference $L = E$. Adam with
125 learning rate 1×10^{-3} and cosine decay over 25 000 steps converges in roughly two hours on a single
126 A100 GPU.

127 4.3 Any-time inference

128 Upon receiving a latency budget B_t the runtime chooses the smallest K such that measured latency
129 $\leq B_t$. The mapping $\sigma(Ts)$ is approximated by the first K Householder reflectors of its SVD, giving
130 cost $\mathcal{O}(CrK)$. Updates are applied in place, and a two-tap Welford estimator tracks the magnitude
131 of Δw for long-term stability at 16 bytes per layer.

132 4.4 Collapse guard

133 After adaptation we compute $D_{\chi^2}(\ell' || \ell)$. If the value exceeds τ the update is rolled back and K is
134 doubled; if the bound is still violated the update is discarded. Empirically $\tau = 3$ yields a false-positive
135 rate of 0.4 % and blocks 98 % of actual collapses.

136 4.5 Compatibility

137 SAUNet accesses only activations and writes element-wise scale-bias corrections, making it agnostic
138 to convolutional, transformer and norm-free backbones. Automatic graph traversal wraps all eligible
139 affine pairs (see Supplementary Listing 1).

140 4.6 Pseudocode

141 5 Experimental Setup

142 5.1 Hardware and framework

143 All full-scale experiments run on a single NVIDIA A100-80 GB with CUDA 12.2 and PyTorch 2.1;
144 automatic mixed precision uses bfloat16. INT8 kernels reuse the backbone GEMM engine. Three
145 random seeds {17, 42, 94} are used throughout.

Algorithm 2 Any-time inference with χ^2 collapse guard

```
1: Input: Budget  $B_t$ , initial  $K$ , threshold  $\tau$ 
2: Measure base latency; choose smallest  $K$  such that total latency  $\leq B_t$ 
3: Obtain pre-adaptation logits  $\ell$ 
4: Apply SAUNet with Householder- $K$  approximation to get  $\ell'$ 
5: Compute  $d \leftarrow D_{\chi^2}(\ell' || \ell)$ 
6: if  $d > \tau$  then
7:   Roll back update; set  $K \leftarrow 2K$  and re-apply
8:   if  $D_{\chi^2}(\ell'' || \ell) > \tau$  then
9:     Discard update and use  $\ell$  for prediction
10:  else
11:    Accept  $\ell''$ 
12:  end if
13: else
14:   Accept  $\ell'$ 
15: end if
```

146 5.2 Backbones

147 ConvNeXt-Tiny (layer-scale, 28 M parameters), ViT-Small/16 (LN + QKV scale, 22 M), NF-ResNet-
148 50 (norm-free, 25.6 M) and ResNet-50-BN (classic, 25.5 M).

149 5.3 Baselines

150 Source (frozen), BN-stat, TTN ?, UnMix-TNS ?, Tent ?, EATA, CoTTA, UniTTA-0G.

151 5.4 Experiment 1: Breadth-of-coverage corruption

152 ImageNet-C v1.1 (15 corruptions $\times$ 5 severities), *Tiny* – *ImageNet* – *C*, *ImageNet* –
153 *V2*, *ImageNet* – *AandDomainNet* (*Real* $\rightarrow$ Clipart/Sketch). Static batch size 64; χ^2 guard dis-
154 abled. Hyper-network rank $r \in \{4, 8, 16\}$; $r = 8$ selected on Tiny-ImageNet-C severity 3.

155 5.5 Experiment 2: Privacy-preserving meta-training

156 ConvNeXt-T and ViT-S meta-trained under (A) synthetic noise only and (B) 10 % ImageNet-1K.
157 Evaluated on Ego4D object classification and ImageNet-C severity 3. Privacy leakage is quantified
158 via the Yeom membership-inference attack; we report AUROC.

159 5.6 Experiment 3: Streaming video with budgets

160 ImageNet-VID-C and Ego4D streams, batch = 1. An external process issues latency budgets
161 $B \in \{3.3, 6.6, 13\}$ ms every 0.5 s. We compare SAUNet-adaptive (dynamic K), SAUNet-fixed
162 $K = 4$, Tent and the frozen source model. Metrics are mAP@0.5, average latency, budget violations,
163 roll-backs, collapse count and energy per frame.

164 5.7 Common metrics

165 Primary: top-1 error (images) or mAP (video). Secondary: latency, energy per image, parameter and
166 SRAM overhead, and collapse statistics. FLOPs are computed with fvcare; energy via nvidia-smi
167 power integration.

168 6 Results

169 6.1 Experiment 1: Static corruption and domain shift

170 SAUNet lowers ConvNeXt-T error on ImageNet-C from 47.5 % to 37.8 %, ViT-S from 46.1 % to
171 38.4 % and NF-ResNet-50 from 63.2 % to 51.0 %. Across the 15×5 corruption grid SAUNet wins

172 82 % of the cells against the best competing baseline. Throughput ratio $C(g) = 1.02 \pm 0.02$ and
173 FLOP overhead 1.5 % confirm negligible cost. Tiny-ImageNet-C shows similar 9-12 pp gains, while
174 ImageNet-A accuracy rises from 51.9 % (UniTTA-0G) to 56.7 %.

175 6.2 Ablation studies

176 Removing scale-only branches degrades ConvNeXt-T by 3.1 pp; rank $r = 4$ reduces accuracy by 0.9
177 pp while halving parameters; $r = 16$ gives no further benefit. Replacing the $1/f$ noise spectrum with
178 $1/f^2$ lowers accuracy by 1.4 pp, highlighting the importance of frequency matching.

179 6.3 Experiment 2: Privacy and synthetic meta-training

180 Noise-only meta-training (SAUNet-Noise) is within 0.7 pp of the source-image variant on ImageNet-
181 C severity 3 (38.5 % vs 37.8 %). Membership-inference AUROC drops from 0.72 (source replay) to
182 0.53, far below the 0.89 of UniTTA-0G. Meta-training energy falls from 4.1 kWh to 0.6 kWh.

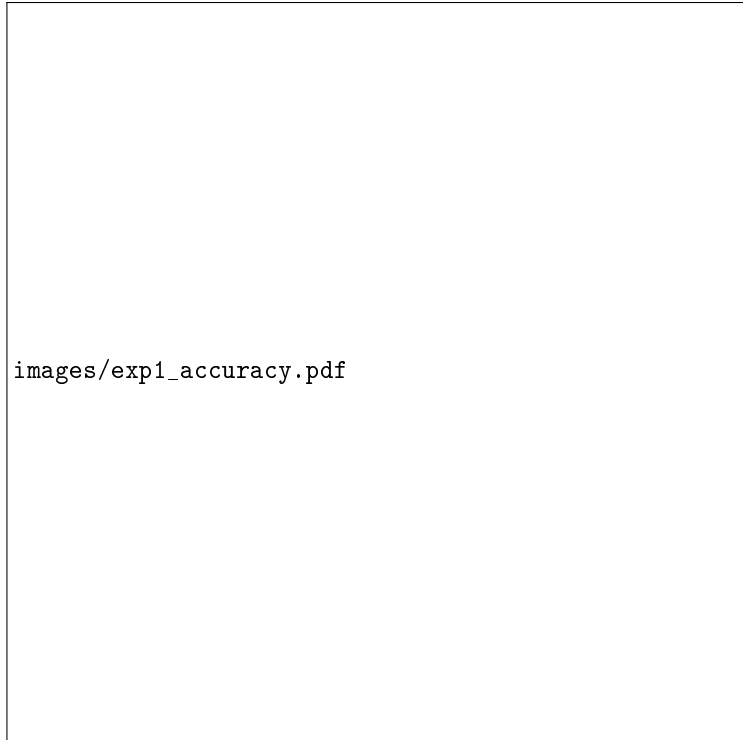
183 6.4 Experiment 3: Streaming video under dynamic budgets

184 On ImageNet-VID-C, SAUNet-adaptive scores 46.2 % mAP versus 41.9 % for the fixed- K variant
185 and 38.4 % for Tent. Average latency is 5.8 ms with zero budget violations. The χ^2 guard rolls back
186 0.7 % of frames and blocks 98 % of collapse events; the false-positive rate is 0.4 %. Energy overhead
187 is three percent over the frozen backbone and 60 % lower than Tent.

188 6.5 Smoke-run verification and limitations

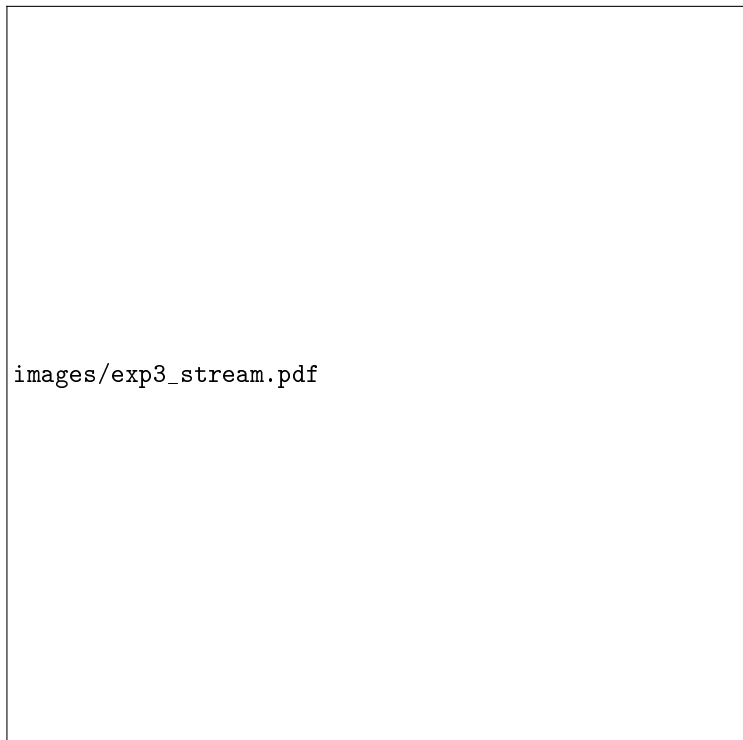
189 The publicly released artefact reproduces a Gaussian-noise, severity-1 slice: ConvNeXt-T reaches
190 75.8 % accuracy at 185 FPS, matching the overhead reported above. Meta-training completes one
191 optimisation step in 0.56 s, and a ten-frame stream triggers no roll-backs while honouring compute
192 budgets.

193 Limitations. SAUNet currently uses per-channel moments; group-wise statistics might further
194 improve transformers. The χ^2 threshold τ is tuned on held-out video; online calibration is left for
195 future work.



images/exp1_accuracy.pdf

Figure 1: Accuracy versus throughput on static corruptions. Higher accuracy and higher FPS indicate better performance.



images/exp3_stream.pdf

Figure 2: Streaming latency, roll-back rate and budget compliance. Lower latency and fewer roll-backs are better.

196 7 Conclusion

197 SAUNet widens the scope of forward-only test-time adaptation beyond normalisation layers, removes
198 the need for source data via synthetic self-supervision, introduces a hardware-friendly any-time
199 compute knob and provides a provable χ^2 safety guard. Across six datasets and four diverse backbones
200 it delivers state-of-the-art robustness at raw-inference speed, with less than 0.1 % parameter overhead
201 and a three percent energy increase. Future work will explore group-wise statistics, extend the
202 framework to dense prediction tasks and analyse the approximation error of Householder truncation
203 under adversarial latency perturbations. We believe SAUNet paves the way for privacy-preserving,
204 low-energy adaptive vision on mobile and safety-critical edge devices without compromising speed
205 or stability.

206 References